



# **Enterprise User Management & Security Administration with Group Policy (GPO)**

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# Introduction

In enterprise environments, System Administrators manage hundreds or even thousands of users through Group Policy Objects (GPO), centralized file sharing, security policies, and Active Directory administration. Rather than configuring each computer individually, administrators apply centralized policies that improve security, consistency, and operational efficiency.

## Objective

To configure Group Policy Objects (GPO), manage Organizational Units (OUs), configure enterprise file sharing with NTFS permissions, implement password policies, map network drives, and verify centralized policy enforcement within an Active Directory environment.

## Theory Topics

### Enterprise Administration

- Enterprise IT Infrastructure

A complex and dynamic ecosystem, comprising various components that work in tandem to support and drive business operations. These components include: Hardware (Servers), Software (OS), Networking & connectivity, End-user devices.

- Centralized Administration

To allows devices, applications, and organizational specifications to be controlled and managed from a central platform.

- Domain Controller Responsibilities
  1. Group Policy Implementation
  2. Validation and Authentication
  3. Permission and Access Regulation
- Best Practices for Domain Administration
  1. Use Multi-Factor Authentication (MFA)
  2. Limit membership
  3. Separate, privileged account for server administration and Active Directory operations
  4. Separate account for unprivileged access (email, browsing, etc)

## Group Policy Objects (GPO) & Management

- What is Group Policy?  
A feature of Windows that provides centralized management and configuration of operating systems, applications, and user settings.
- What is GPO?  
A virtual collection of policy settings with a unique name.
- Local Policy([gpedit.msc](#)) vs Domain Group Policy

| Local Group Policy                                                                                                   | Domain Group Policy                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Local policies are applied to an individual computer.                                                                | Domain Group Policies are applied to multiple computers and users within a network.                                        |
| These policies are managed through the Local Group Policy Editor ( <a href="#">gpedit.msc</a> ).                     | These policies are managed centrally using the Group Policy Management Console (GPMC) in AD.                               |
| These policies are stored on the local computer and are separate from AD.                                            | These policies are created and applied at the domain level.                                                                |
| These policies are suitable for stand-alone computers or environments where centralized management is not necessary. | These policies are ideal for larger organizations where policies need to be consistently enforced across numerous systems. |

- Group Policy Processing

*Group Policy Objects (GPO) are processed in the following order:*

1. The local GPO is applied.
  2. GPOs linked to sites are applied.
  3. GPOs linked to domains are applied.
  4. GPOs linked to organizational units (OUs) are applied. In a nested OU structure, GPOs linked to parent OUs are applied first, followed by GPOs linked to the child OUs.
- User Configuration vs Computer Configuration
    - User settings of a GPO will only affect user accounts that reside in the OU(s) that are in the scope of where that GPO is linked.
    - Computer settings will only affect computer accounts that reside in the OU(s) that are in the scope of where that GPO is linked.
  - Group Policy Inheritance

AD feature where child containers (like Organizational Units or domains) automatically receive the Group Policy Object (GPO) settings applied to their parent containers.

**Enforced** (formerly called "No Override") : ensures a parent GPO always applies to child containers.

**Block** Inheritance : stops an Organizational Unit (OU) from receiving policies from its parent containers.

- Linking a GPO

Right-click YourDomainName, and then click Link an Existing GPO.

- Updating Policies

`gpupdate /force`

- Verifying Policy Application

`gpresult /r`

## **Password & Account Security**

- Security Best Practices

a minimum length of 16 characters, complete uniqueness per account, a mix of character types, and the use of multi-factor authentication managed through a trusted digital vault.

## **File Server Administration**

- Shared Folders

Shared Folders is the component in a network where users from various hosts can access a certain folder or files over the network. Shared Folders make it easy to share files over the network.

- NTFS Permissions

*found in security tab*

*NTFS (NT File System) stands for New Technology File System (NTFS).*

NTFS permissions are used to manage access to the files and folders that are stored in NTFS file systems.

1. **Full control:** Allows users to read, write, change, and delete files and subfolders. In addition, users can change permissions settings for all files and subdirectories.
2. **Modify:** Allows users to read and write of files and subfolders; also allows deletion of the folder.
3. **Read & execute:** Allows users to view and run executable files, including scripts.
4. **List folder contents:** Permits viewing and listing of files and subfolders as well as executing of files; inherited by folders only.
5. **Read:** Allows users to view the folder and subfolder contents.
6. **Write:** Allows users to add files and subfolders, allows you to write to a file.

- Share Permissions

*found in sharing tab*

manage access to folders shared over a network, such as a shared folder located on a centrally hosted file server. Share permissions do not apply to users who log on locally. Share permissions apply to all files and folders in the share; you cannot granularity control access to sub-folders or objects on a share. You can specify the number of users who are allowed to access the shared folder.

| Feature                   | Share Permissions                                  | NTFS Permissions                                      |
|---------------------------|----------------------------------------------------|-------------------------------------------------------|
| Ease of Management        | Easy to apply and manage                           | More granular control of folder and contents          |
| File System Compatibility | Works with both NTFS and FAT or FAT32 file systems | Available for NTFS file systems only                  |
| Connection Restrictions   | Can limit concurrent connections                   | Cannot limit connections                              |
| Configuration Location    | Advanced Sharing > Permissions                     | File/Folder Properties > Security tab                 |
| Scope of Access Control   | Regulates remote network access only               | Secures both local and remote access                  |
| Local Access Impact       | Does not affect local access                       | Controls local and remote access                      |
| Security Coverage         | Limited to networked environments                  | Secures both local and remote access                  |
| Complementary Use         | Requires NTFS permissions for full protection      | Functions independently but enhances overall security |

- Effective Permissions

The actual, resulting access rights a user or group has over a Group Policy Object (GPO), calculated by combining explicit permissions, inherited permissions from Active Directory, and nested group memberships.

## **Network Drive Mapping**

- Network Shares

A way to make files and folders available over a network.

- Drive Mapping

The process of associating a local drive letter with a shared storage area on a remote computer (often called a file server) over a computer network. Once a drive has been mapped, applications on the client computer can read and write files on the remote storage as if it were a locally attached hard disk drive, using the assigned letter in file paths

- UNC Path

Universal naming convention (UNC) paths, which are used to access network resources.

A server or host name, which is prefaced by \\. The server name can be a NetBIOS machine name or an IP/FQDN address (IPv4 as well as v6 are supported).

A share name, which is separated from the host name by \. Together, the server and share name make up the volume.

- Persistent Network Drives

a remote shared folder on a server or network attached storage (NAS) that is mapped to a local drive letter on a computer and configured to automatically reconnect every time the system restarts or the user logs back in.

“Reconnect at sign-in”

## Basic Troubleshooting

- GPO not Applying —> `gpupdate /force`
- Permission Issues —> Match NTFS and Share Permissions / Check Effective Access (in advanced settings)
- Folder Sharing Problems —> Verify Network Discovery (file & printer sharing)/ test UNC path
- User Authentication Issues —> Re-authenticate Session: Clear active connections on the client command line using `net use * /delete` before trying to map the network drive again.
- Policy Replication —> verifying that both the Active Directory Group Policy Container (GPC) and the SYSVOL Group Policy Template (GPT) match across all domain controllers using built-in replication tools and the Group Policy Management Console.

## Software Requirements

**Software:** VM workstation

**Environment1:** Windows Server 2019

**Environment2:** Windows 10 Client VMs

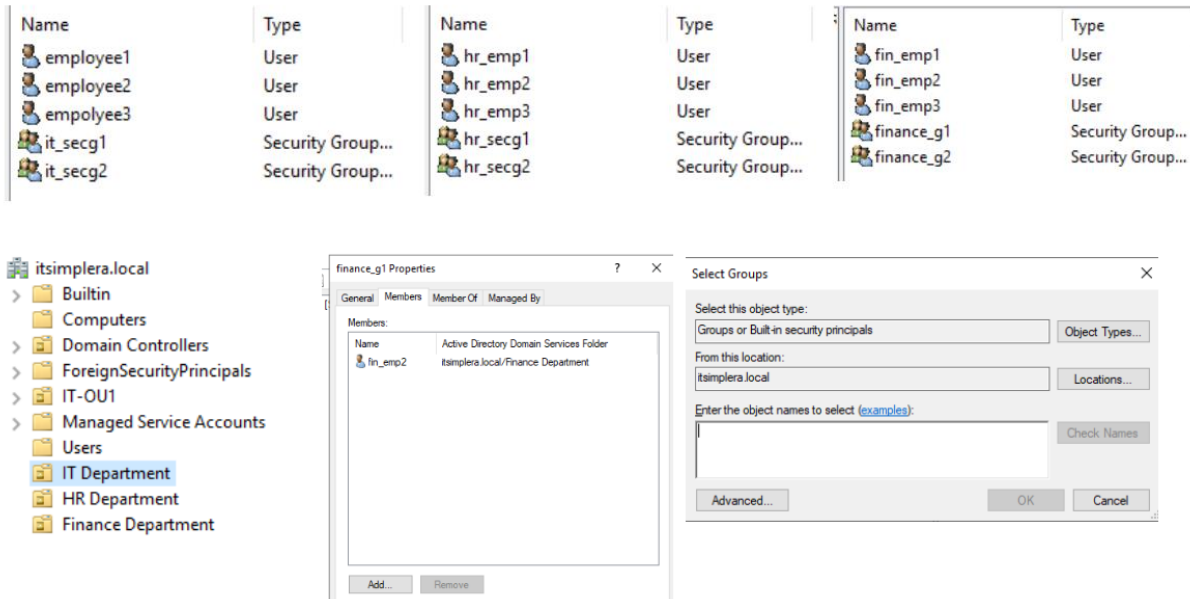
**Prerequisites:** AD DS, DNS & DHCP configured

**Access Level:** Administrator Account

# Implementation

## 1. Creating an Enterprise Organizational Structure

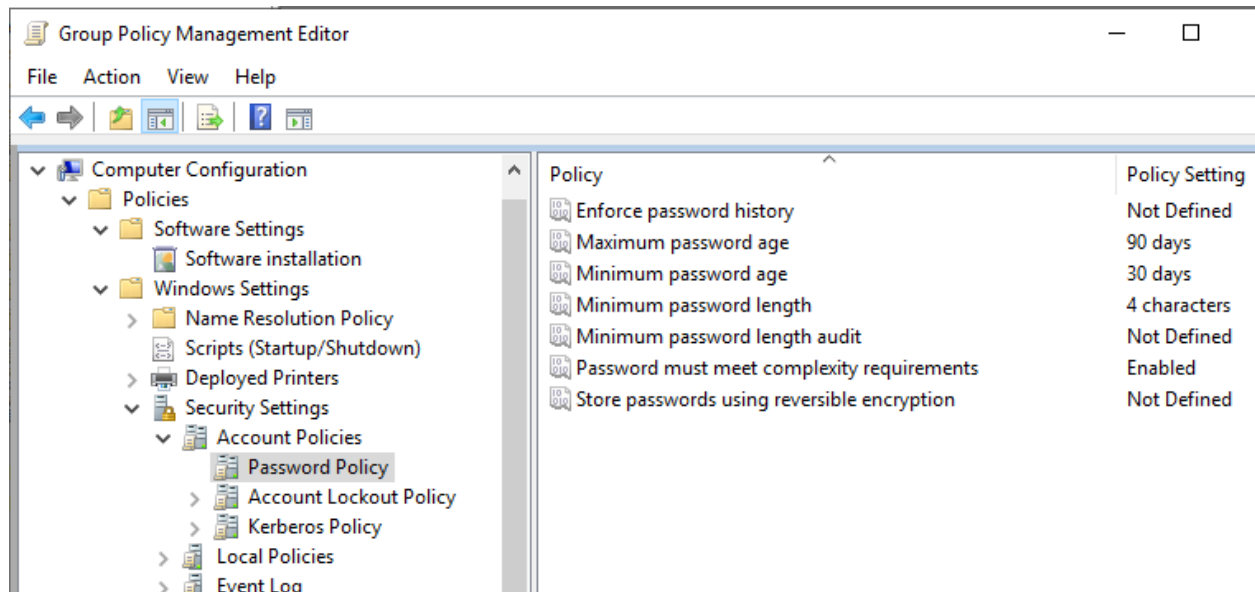
3 OUs each with 3 users and 2 security groups



## 2. Configuring Password & Account Policies

Using Group Policy Management (gpmc.msc)

right-click on Default Domain Policy -> edit -> computer configuration -> policies -> windows policies -> security settings -> account policies



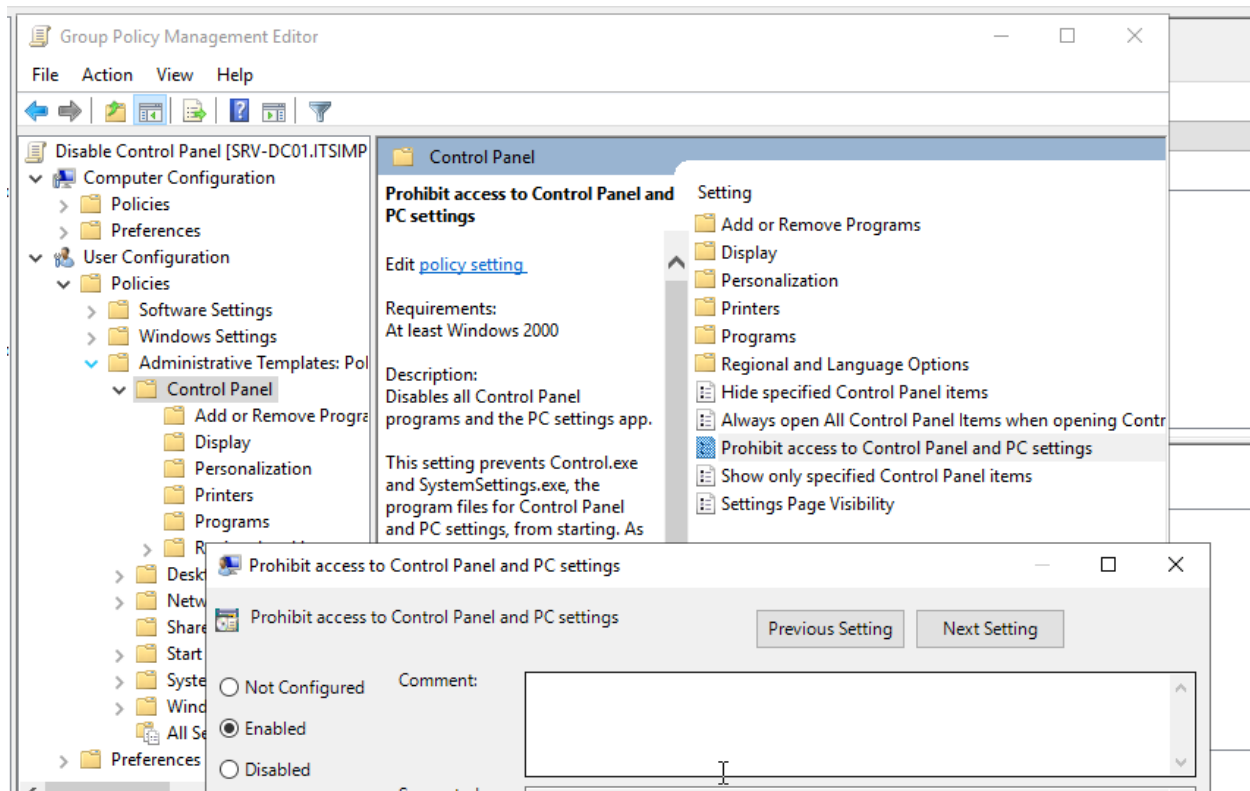
### 3. Creating & Applying Group Policy Objects (GPO)

Using Group Policy Management (gpmc.msc)

right-click on Group Policy Objects -> new

right-click on the created object -> edit -> user configuration -> policies -> administrative templates ->

- control panel -> prohibit access to control panel
- System -> prevent access to the command prompt
- System -> removable storage access -> removable disks -> deny read/write access
- Desktop -> desktop -> desktop wallpaper
- Start menu and task bar -> remove run menu



\_ gpupdate /force : Forces an immediate refresh of all Group Policy settings on the computer.

```
C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.
```

\_ gpresult /r : Generates a Group Policy Result summary report in the command line (the /r stands for Result). It shows a list of all the Group Policy Objects that have successfully applied to the current user and computer.

```

Applied Group Policy Objects
-----
    Default Domain Controllers Policy
    Default Domain Policy

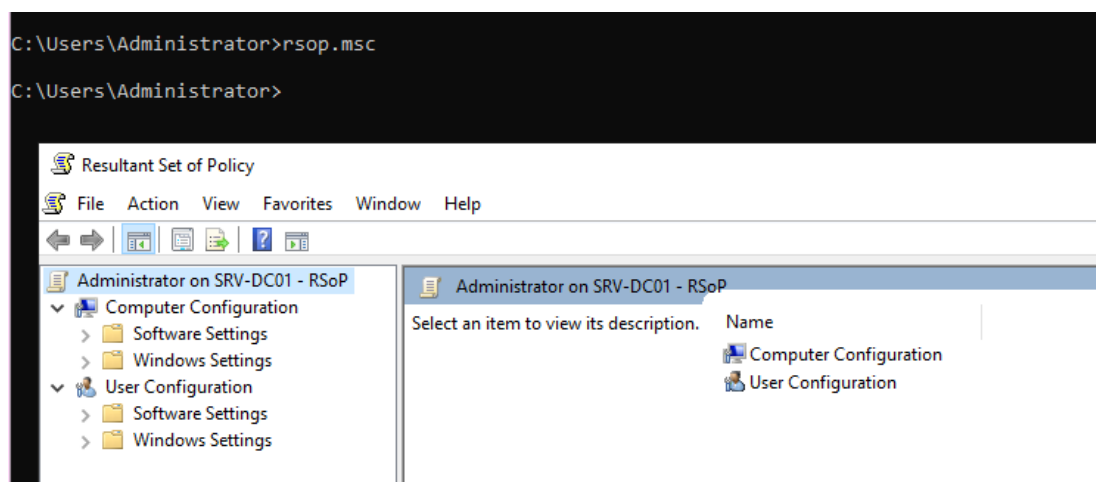
The following GPOs were not applied because they were filtered out
-----
    Local Group Policy
        Filtering:  Not Applied (Empty)

The computer is a part of the following security groups
-----
    BUILTIN\Administrators
    Everyone
    BUILTIN\Pre-Windows 2000 Compatible Access
    BUILTIN\Users
    Windows Authorization Access Group
    NT AUTHORITY\NETWORK
    NT AUTHORITY\Authenticated Users
    This Organization
    SRV-DC01$
    Domain Controllers
    NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS
    Authentication authority asserted identity
    Denied RODC Password Replication Group
    System Mandatory Level

USER SETTINGS
-----
    CN=Administrator,CN=Users,DC=itsimplera,DC=local
    Last time Group Policy was applied: 8/3/2026 at 11:58:43 AM
    Group Policy was applied from:      SRV-DC01_itsimplera_local

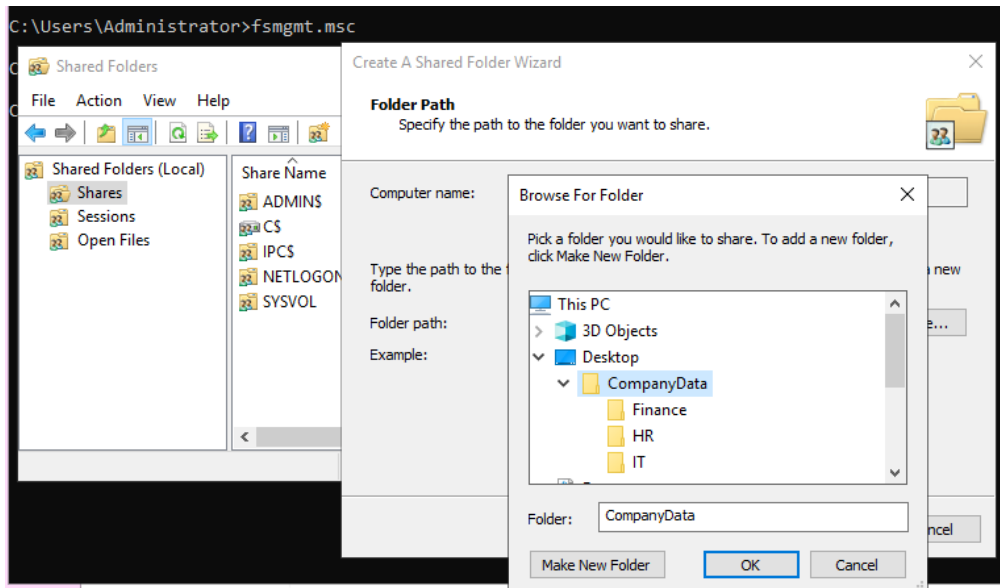
```

\_ rsop.msc : Launches the Resultant Set of Policy graphical console tool. It opens a window that visually maps out all the policies currently enforced on the machine.



## 4. Configuring Enterprise File Sharing

- a. Create regular folders on the server
- b. fsmgmt.msc -> shares -> right-click on empty space -> new share -> choose folder (all users read-only access)



### Sharing was Successful

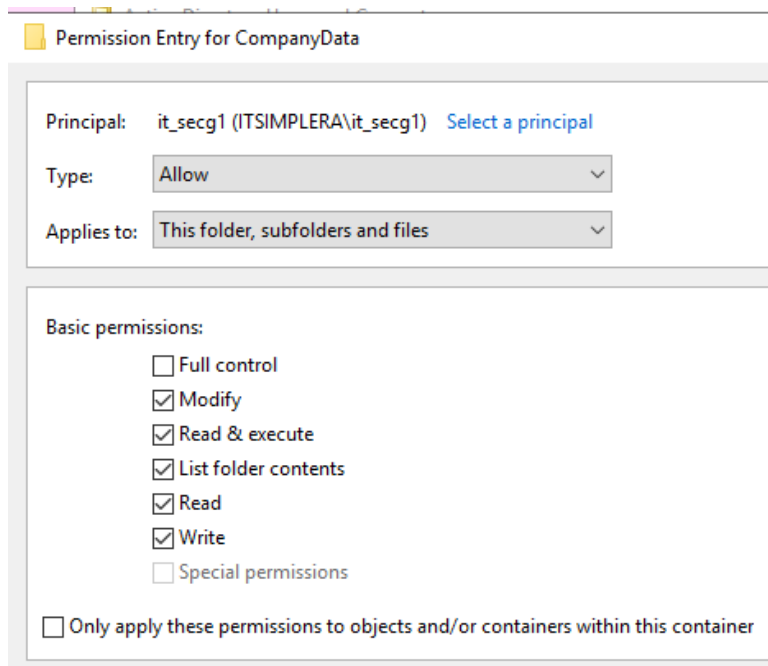
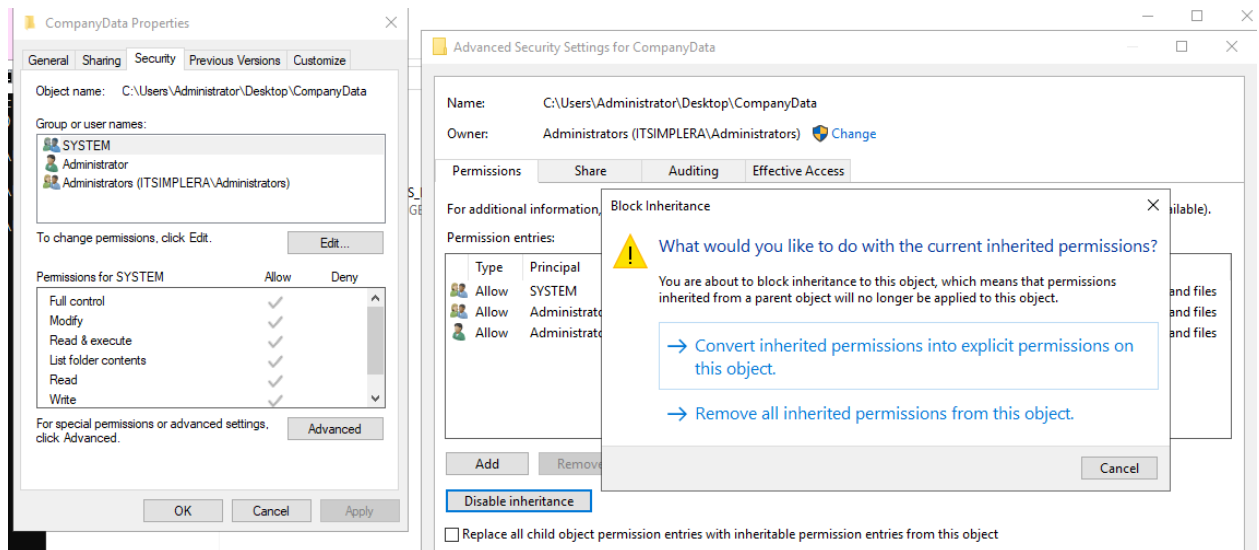
#### Status:

You have successfully completed the Share a Folder Wizard.

#### Summary:

You have selected the following share settings on \\\SRV-DC01:  
Folder path: C:\Users\Administrator\Desktop  
CompanyData  
Share name: CompanyData  
Share path: \\\SRV-DC01\CompanyData

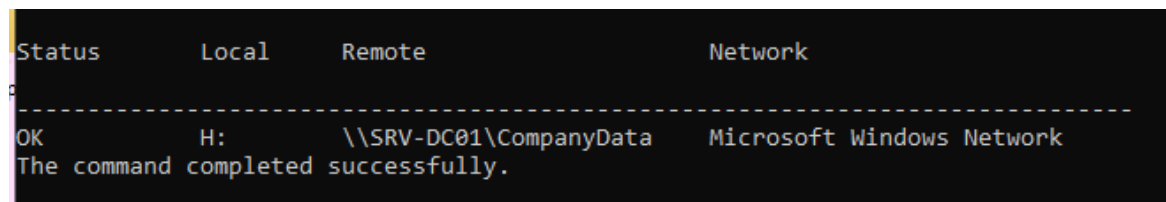
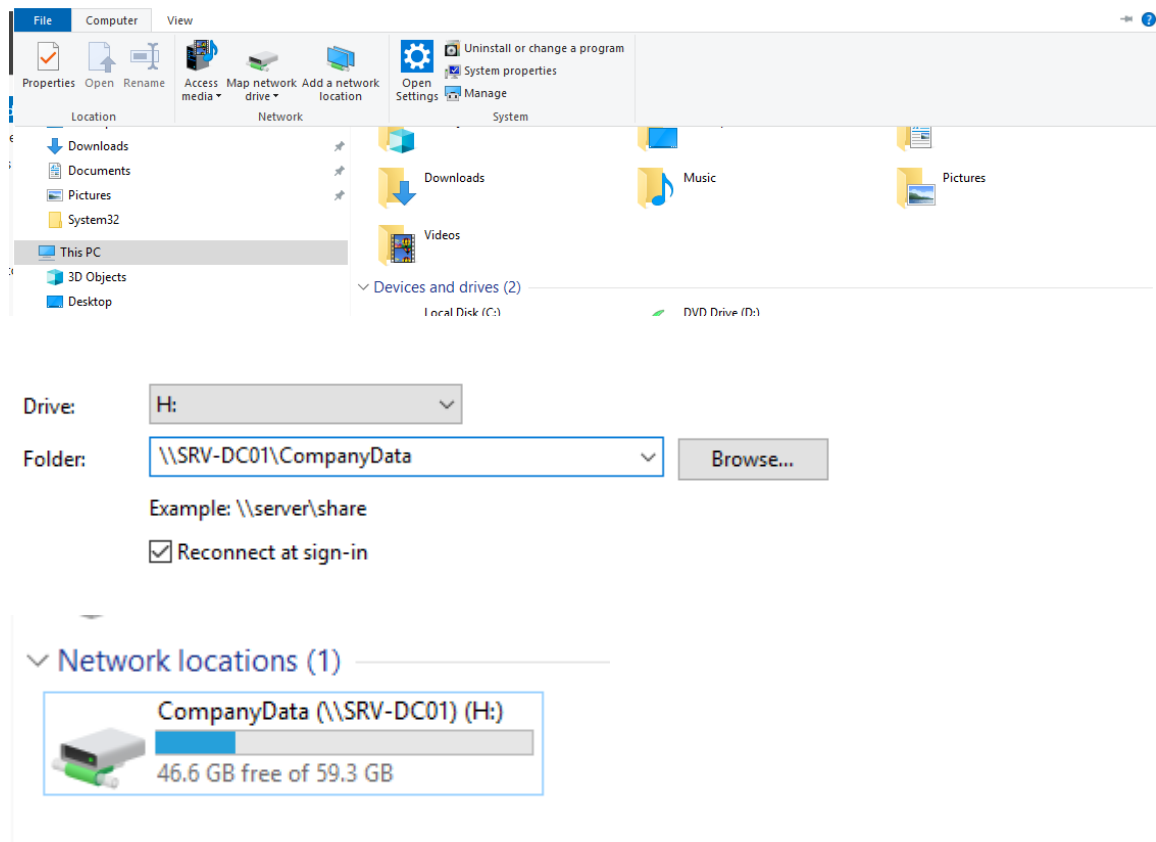
- c. right-click on shared folder -> properties -> security -> advanced -> disable inheritance (to convert inherited permissions into explicit ones) -> add AD security groups and grant access



## 5. Mapping Network Drives

Access 'This Computer' -> Computer Tab -> map network drive -  
> choose letter & specify shared folder as

“\\Server(Computer)Name\\SharedFolderName” -> dfs



## 6. Some Routine AD Tasks

dsa.msc

- Reset Password: right-click on user -> reset password
- Unlock a Locked Account: right-click on user -> properties -> account tab -> unlock account check box
- Disable a User: right-click on user -> disable account | -> properties -> account tab -> account is disabled check box

- Enable the User: right-click on user -> enable account
- Move User between OUs: right-click on user -> move
- Delete a User: right-click on user -> move

## 7. Troubleshooting

**Problem#1:** Map Drive Disconnected

**Root Cause:** (a) Network connection disabled

(b) Wrong server/file name

(c) Folder is unshared

**Fix:**

(a) Enable network adapter (ethernet)

(b) Correct the server or folder name path in the Folder field

(c) Right-click the folder —> Properties —> Sharing tab —> Advanced Sharing—> Share this folder

**Problem#2:** User cannot access Shared Folder.

**Root Cause:** (a) Folder not shared

(b) User not added to the shared folder

**Fix:**

(a) Right-click the folder —> Properties —> Sharing tab —> Advanced Sharing—> Share this folder

(b) Right-click the folder —> Properties —> Sharing tab —> Advanced Sharing —> Permissions —> Add —> Grant permissions.

# Learning outcomes

Understanding enterprise Active Directory administration, implementing and managing Organizational Units (OUs), users, and security groups, configuring and managing Group Policy Objects (GPOs), implementing password and account security policies, configuring secure file sharing using NTFS permissions, mapping network drives for domain users, troubleshooting common Active Directory and Group Policy issues, and developing the practical skills required to perform the essential day-to-day responsibilities of a Windows System Administrator.

## Conclusion

Centralized administration is essential for maintaining secure, consistent, and efficient enterprise IT environments. Technologies such as Active Directory and Group Policy enable administrators to manage users, enforce security policies, and control access to resources from a single location, simplifying administration while improving security and reliability.

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